

D5 — Retrospective & Self-Assessment

WorkflowScript — CSE341 Course Project, Part 1

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What is working

The core Part 1 pipeline is fully functional. The lexer correctly tokenises all WorkflowScript source files, tracks line numbers on every token, and reports illegal characters with their position. The recursive-descent parser handles all top-level declarations (`task`, `pipeline`, `fun`), all statement forms (`var`, `if`, `while`, `return`, `run`, assignment), and the full operator-precedence hierarchy from `||` down to unary `- !`. The `--dump-ast` flag produces a readable tree printout, error messages always include a line number, and the exit-code contract (0 = success, 1 = error) is enforced. All three valid example programs parse cleanly and all five malformed programs produce the expected error messages.

What was hard

The trickiest implementation detail was field access on `task` values. The fields `timeout`, `retries`, and `parallel` are reserved keywords in the token package, so after a `.` the parser received a `KW_TIMEOUT` token rather than an `IDENT`. The initial `parsePrimary` logic only accepted `IDENT` as a field name and silently failed on `build.timeout`. Fixing this required extending the field-access branch to also accept keyword tokens and use their lexeme as the field name. This was a good reminder that keyword/identifier boundaries in a lexer have downstream consequences that are not always obvious at design time.

What I plan to fix in Part 2

The type checker and interpreter are already implemented ahead of schedule, so Part 2 work will focus on the design document rather than new code. Specifically, I plan to complete the Semantics (§4.4), Type System (§4.5), Expressions & Assignment (§4.7), and Design Rationale (§4.8) sections of D1, and update the D3 test report to include actual interpreter output and a type-error example. I also want to revisit the `task` field-ordering constraint — either relaxing it in the parser or improving the error message — and add at least one domain-specific runtime error beyond division by zero.